Daily Requirements

15-20% of total calorie requirements should be from fat

Sources

Animal sources: *Fish, egg, meat, milk and milk products.*

Plant sources: *oil seeds (Groundnut, mustard, cotton seed and coconut oil), nuts.*



Functions

- Supplies energy (9 kcals/ gm)
- Improve the palatability of food (flavor and taste)
- Supports body organs like liver and kidneys
- Provides insulation and thermoregulation against cold
- Provides essential fatty acids which helps in growth, promotion and maintenance of skin integrity
- Helps in formation of hormones in the body
- Helps in transportation of fat soluble vitamins

Digestion and absorption

There is no digestion of fats in mouth and very little in stomach. In the small intestine, presence of fat stimulates cholecystokinin and secretin which further stimulates pancreatic juice and bile. Food mixes with bile and emulsified. Pancreatic lipase hydrolyses and yields fatty acids and monoglycerides. These pass to small intestine and combined with intestinal lumen and absorbed in to blood stream.



Excessive accumulation of fat causes obesity in all age groups.

Obesity increases the risk of following diseases:

- Diabetes mellitus
- High blood pressure
- Stroke
- Gall stones
- Coronary artery diseases
- Low back pain and osteoarthritis



STUDENT'S ACTIVITY

For the following nutrients, can you say why they are important and name one source?

Carbohydrates

_____

Proteins

_____

Fats and oils

_____

IV. Vitamins

vitamins are complex chemical substances required by body in very small amounts. Vitamins in food are for the protection and regulation of body functions.

- **Vitamins** are divided in to two major groups
- Fat soluble vitamins- Vitamin-A, D, E and K
- Water soluble vitamins- Vitamins of B group and vitamin C

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